

Sports intelligence - Teacher Guide

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QRU Sports Intelligence Lab(TM) Teacher Guide

CHAPTER 1

Train Your Brain Like an Athlete Trains Their Body(TM)

CHAPTER 2

1. Guide Purpose

QRU Sports Intelligence Lab(TM) helps learners see that difficult things can be learned. It turns math learning into an athlete-style training experience where children practice thinking through sports, strategy, teamwork, repetition, feedback, and reflection.

The goal is not only correct answers.

The goal is developing:

- Confidence
- Curiosity
- Persistence
- Reasoning
- Communication
- Self-awareness
- Problem solving
- Lifelong learning ability

Core message:

"Champions train their bodies.

Brain Athletes train their minds."

CHAPTER 3

2. Core QRU Principle

"Children do not need more information.

They need stronger understanding systems."

In this guide, mathematics is taught as a mental sport. Learners are not treated as "good at math" or "bad at math." They are treated as Brain Athletes who can improve through meaningful practice.

CHAPTER 4

3. Teacher Role

The teacher becomes a coach.

The teacher serves as:

- Coach
- Facilitator
- Encourager
- Guide

The purpose is not to pressure learners into perfection. The purpose is to help them discover:

"I can learn difficult things."

CHAPTER 5

4. Learning Frame: What Is Sports Intelligence?

Sports intelligence in this guide means using sports as a doorway into stronger thinking.

Learners connect mathematics to:

- Sports
- Strategy
- Competition
- Practice
- Teamwork
- Decision-making
- Real-world problem solving

The learner becomes a Brain Athlete who trains the mind through practice, reflection, and growth.

CHAPTER 6

5. Key Vocabulary

6.1 Multiplication

Multi = many

Plication = groups

Meaning:

Many groups combined.

Multiplication is not just memorizing facts. It is understanding how many equal groups combine to make a total.

6.2 Intelligence

The ability to learn, understand, adapt, and solve problems.

CHAPTER 7

6. Main Teaching Analogy

Use this analogy throughout the lesson:

| Sports Training | Brain Training |

|---|---|

| Brain = Muscle | The mind strengthens through practice |

| Practice = Training | Thinking improves through repetition |

| Mistakes = Training Reps | Mistakes help learners adjust |

| Knowledge = Equipment | Knowledge gives learners tools |

| Understanding = Game Strategy | Understanding helps learners know what to do |

| Reflection = Game Film Review | Reflection helps learners notice patterns and improve |

Teacher language:

"Athletes do not become stronger by doing everything perfectly the

first time. They practice, get feedback, adjust, and try again. Brain Athletes train the same way."

CHAPTER 8

7. Emotional Safety Standard

This learning model protects dignity.

Multiplication is not:

- Memorizing without understanding
- Racing against others
- Measuring intelligence

When a learner says:

"I'm bad at math."

Use the QRU response:

"You may need more practice, different explanations, or a different strategy."

This keeps the focus on growth, not shame.

CHAPTER 9

8. Lesson Flow: The QRU Translation(TM) Cycle

Use the following learning cycle:

9.1 1. Question

Begin with curiosity.

Ask:

"What is multiplication?"

"How can multiplication help athletes?"

"How does understanding patterns improve decisions?"

9.2 2. Understanding

Teach the meaning:

Multiplication means many groups combined.

Connect the idea to sports:

"If an athlete repeats the same action across practices, games, or quarters, multiplication helps us find the total."

9.3 3. Practice

Use repeated sports examples.

Example:

A basketball player makes 8 free throws each practice.

The player practices 5 times.

$8 \times 5 = 40$ free throws.

9.4 4. Connection

Ask learners to connect math to sports strategy.

Prompt:

"How does knowing the total help the athlete understand progress?"

9.5 5. Application

Learners create their own sports problem.

Activity:

- Choose a sport.
- Choose a repeated action.
- Choose how many times it happens.
- Write a multiplication sentence.
- Solve it.
- Explain the strategy.

9.6 6. Reflection

Use the "game film review" analogy.

Ask:

"What did you understand better after practice?"

"What mistake helped you learn?"

"What strategy would you use next time?"

9.7 7. Growth

Close with the identity message:

"Every Brain Athlete begins as a rookie. Champions are created through intentional practice."

9. Core Lesson: Multiplication Sports Lab

10.1 Lesson Title

Multiplication as Many Groups Combined

10.2 Teacher Opening

"Today we are going to train our brains like athletes train their bodies. Athletes improve through repetition, feedback, adjustment, and reflection. Learning works in a similar way. In QRU Sports Intelligence Lab(TM), math becomes a mental sport."

10.3 Teach the Concept

Multiplication means:

Many groups combined.

Use a sports example:

"If a quarterback completes 5 passes each quarter and there are 4 quarters, we can find the total number of completed passes."

Write:

$$5 \times 4 = 20 \text{ passes}$$

Explain:

- 5 passes in each quarter
- 4 quarters
- Many equal groups combined
- Total: 20 passes

10.4 Second Example

A basketball player makes 8 free throws each practice.

The player practices 5 times.

$8 \times 5 = 40$ free throws.

Explain:

- 8 free throws in each practice
 - 5 practices
 - Many equal groups combined
 - Total: 40 free throws
-

CHAPTER 11

10. Guided Practice Questions

Use the verified examples.

11.1 Football Question

A quarterback completes 5 passes each quarter. There are 4 quarters.

Ask:

1. What is the repeated group?
2. How many groups are there?
3. What multiplication sentence can we write?
4. What is the total?

Answer:

$5 \times 4 = 20$ passes

11.2 Basketball Question

A player makes 8 free throws each practice. The player practices 5 times.

Ask:

1. What is the repeated group?
2. How many practices are there?
3. What multiplication sentence can we write?
4. What is the total?

Answer:

$$8 \times 5 = 40 \text{ free throws}$$

CHAPTER 12

11. Student Activity: Design Your Own Sports Problem

Learners create their own multiplication problem using sports.

Steps:

1. Pick a sport.
2. Pick an action that can repeat.
3. Decide how many are in each group.
4. Decide how many groups there are.
5. Write the multiplication sentence.
6. Solve the total.
7. Explain the strategy.
8. Reflect on improvement.

Reflection prompt:

"How did creating your own problem help you understand multiplication?"

CHAPTER 13

12. Strategy Talk: Understanding Builds Confidence

After practice, guide learners into reasoning.

Ask:

- "How did multiplication help us count faster?"
- "How does understanding groups help athletes track progress?"
- "How can patterns help with decisions?"
- "What did you do when the problem felt difficult?"

Memory sentence:

"Understanding creates confidence."

Teacher emphasis:

Correct answers matter, but understanding matters too. A learner who can explain a strategy is building stronger thinking.

CHAPTER 14

13. Quiz Bank

14.1 Recall Questions

1. What does multiplication mean?

- Answer: Many groups combined.

2. What does intelligence mean?

- Answer: The ability to learn, understand, adapt, and solve problems.

3. What is a Brain Athlete?

- Answer: A learner who trains the mind through practice, understanding, and growth.

14.2 Application Questions

1. A quarterback completes 5 passes each quarter for 4 quarters. What is the total?

- Answer: $5 \times 4 = 20$ passes.

2. A basketball player makes 8 free throws each practice for 5 practices. What is the total?

- Answer: $8 \times 5 = 40$ free throws.

14.3 Reasoning Questions

1. Why is multiplication useful in sports?

- Answer: It helps combine repeated groups and understand totals.

2. How does understanding patterns improve decisions?

- Answer: Patterns help learners notice what is happening and choose a better strategy.

14.4 Reflection Questions

1. What did you understand better after today's practice?
 2. What mistake became a training rep for your brain?
 3. What strategy helped you most?
 4. What would you try next time?
-

CHAPTER 15

14. Common Misunderstanding and Teacher Response

15.1 Myth

"I'm bad at math."

15.2 QRU Understanding

"You may need more practice, different explanations, or a different strategy."

15.3 Teacher Guidance

Do not label the learner. Reframe the challenge as training.

Say:

"Your brain is still training. Let's try another strategy."

CHAPTER 16

15. Reflection: Game Film Review

End the lesson by comparing reflection to watching game film.

Teacher language:

"Athletes review what happened, notice patterns, adjust their strategy, and come back stronger. Brain Athletes do the same thing with learning."

Reflection questions:

- What happened during practice?
 - What pattern did you notice?
 - What did you adjust?
 - What helped you improve?
 - What will you practice next?
-

CHAPTER 17

16. Extension Activities

Use these activities to deepen understanding:

17.1 Create a Training Plan

Learners make a simple plan for practicing a math skill.

Prompt:

"What will you practice? How will you know you are improving?"

17.2 Explain Your Strategy

Learners solve a sports multiplication problem and explain how they thought through it.

Prompt:

"Do not only give the answer. Explain the thinking."

17.3 Teach Someone Else

Learners explain multiplication as "many groups combined" to another person.

Teaching others helps multiply knowledge.

CHAPTER 18

17. Use Settings

QRU Sports Intelligence Lab(TM) can be used in:

- Homeschool
- Classrooms
- Family learning
- Coaching programs
- Educational products

Caregivers can also act as coaches by encouraging practice, reflection, and confidence.

CHAPTER 19

18. Review Requirements

Before using or sharing a lesson, review for:

- Mathematical accuracy
- Age appropriateness
- Educational alignment
- Emotional safety

The lesson should help learners feel capable, not judged.

CHAPTER 20

19. Closing Message for Learners

"Every Brain Athlete begins as a rookie. You train, practice, struggle, improve, and grow. Champions are not created by talent alone. Champions are created through intentional practice."

Final memory sentence:

"Champions train their bodies.

Brain Athletes train their minds."

Continue your understanding at QRU:

